ENROLLMENT PERFORMANCE MONITORING & DEMAND FORECASTING

JANUARY 2024 – NOVEMBER 2025

Presented by Suciningtyas Nur Alifah

Sales Data Scientist & Analyst

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WORK PLAN AND WEEKLY PROGRESS REPORT

1. Dibimbing memberikan Template Activity Log untuk masing-masing Peserta Internship pada saat Internship Briefing atau F
2. Peserta diwajibkan untuk melengkapi "Data Peserta" di
3. Peserta melakukan pengisian Activity Log secara rutin s intership. Selain itu, jika ditemui kendala yang perlu disele
4. Pengisian Activity Log dilakukan setiap hari Sabtu dan M pelaksaan program internship mulai senin di minggu yang
5. Jika peserta tidak mengisi Activity Log dua minggu bertu Internship Program).
6. Untuk penjelasan pengisian activity log:
- **Detail yang dikerjakan:** Peserta diminta untuk menjelas
- **Output yang dikerjakan:** Peserta diminta untuk menjela
- **Kendala dari pekerjaan:** Peserta diminta untuk menjela pribadi ataupun kendala dari luar (eksternal).
- **Catatan tambahan**: Bisa tambahkan link penunjang deta

(READ ME) Guideline Pengisian Logbook | (READ ME) Contoh Act

PROJECT TITLE	Sales Data Scientist & Analyst
COMPANY NAME	Dibimbing.com
PERIOD	Rabu, 17 Desember 2025 - Sabtu, 28 Februari 2026

TASK ID	TASK TITLE	Note	START DATE	DUE DATE	DURAT IN DA
1	Preparations				
1.1	Set project timelines and milestones	Create a schedule and targ	5 Jan 2026	7 Jan 2026	2
1.3	Data Understanding	Examine the structure, form	8 Jan 2026	21 Jan 2026	13
1.5	Create PPT	Create template (intro)	15 Jan 2026	27 Jan 2026	12
2	Data Preparations				
2.1	Data Cleaning	Fixing errors and inconsiste	20 Jan 2026	27 Jan 2026	7
2.2	Data Transformation	Aggregate data, create new	25 Jan 2026	27 Jan 2026	2
3	EDA - Identifying Opportunities for Development				

sheet1

WORKFLOW

1. Data Understanding
2. Data Preparation
3. Problem Statement
4. EDA
5. Create Dashboard
6. Model Development
7. Forecast Generation

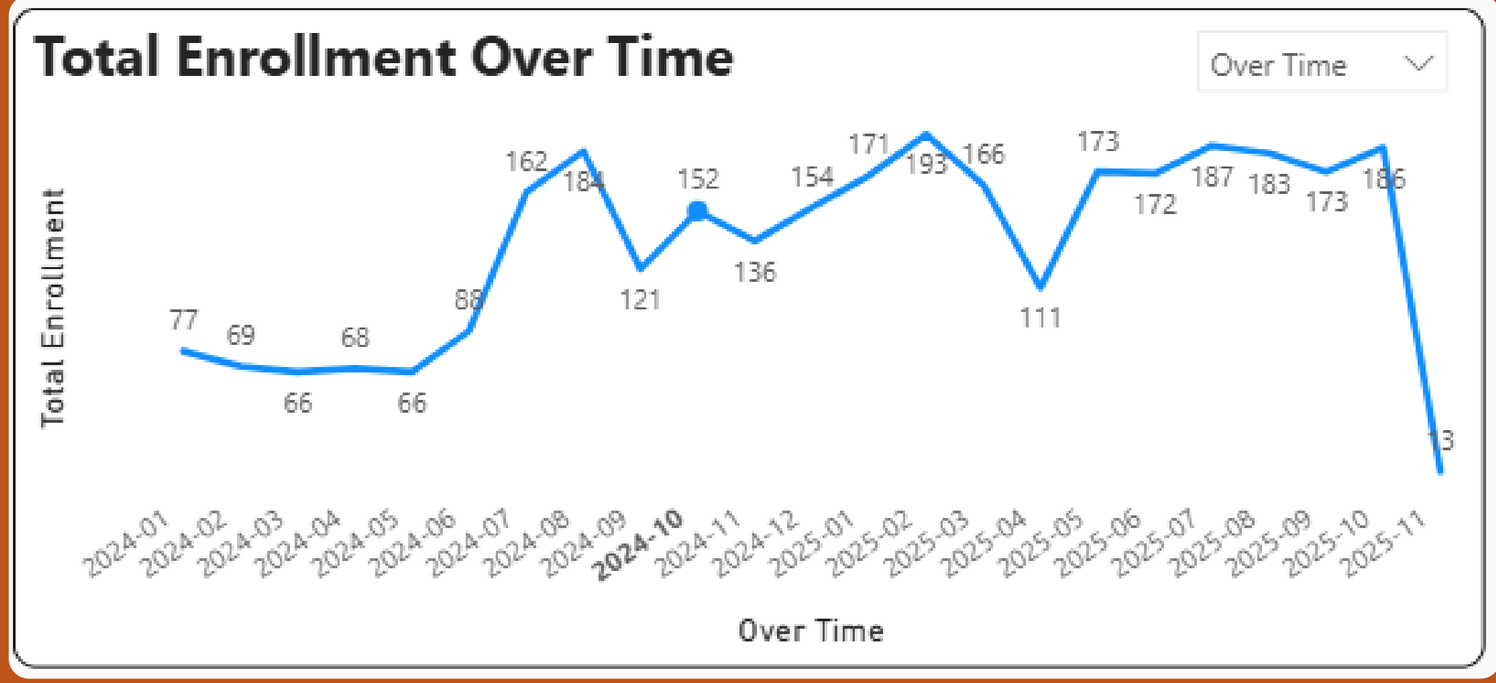
DATA UNDERSTANDING

The dataset consists of **3,071 rows and 40 columns**, representing participant and sales-related information from multiple sources. The dataset spans **January 2024 to October 2025** and covers both online and offline programs. **Each row corresponds to a single participant**, includes Source Table, Date Information, Demographic & Geographic Attributes (Age, Region, Job Category, Education), Marketing (Acquisition Channel), Behavioral (Motivation text)

- Several columns are stored as object/str (e.g., date, age, date) → **format standardization**
- **Time-related columns overlap** with the main date column → **removed**
- Dataset shows no null values, but manual review found empty strings, hyphens (-), and whitespace in several columns → **cleaning and missing value handling**
- Multiple categorical variables contain **inconsistent labels** and mixed data types → **require normalization**
- Several features have **high-cardinality free-text values**, increasing preprocessing complexity → **generate world cloud**

link notebook

PROBLEM STATEMENT



In recent periods, enrollment overall (several products) have shown **declining enrollments across new batches**. **However, this caused by incomplete reporting.**

Data Period

- **Start: 2 January 2024**
- **End: 3 November 2025**
- **Duration: 671 days (~22 months)**

Objectives

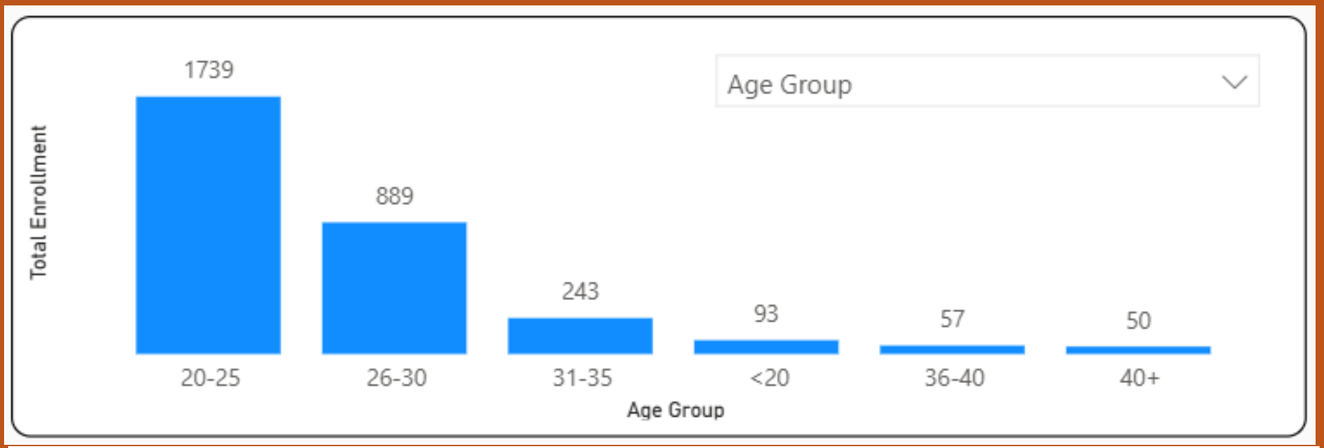
1. Build product-level enrollment monitoring
2. Detect growth & volatility patterns
3. Forecast short-term demand (3 months)

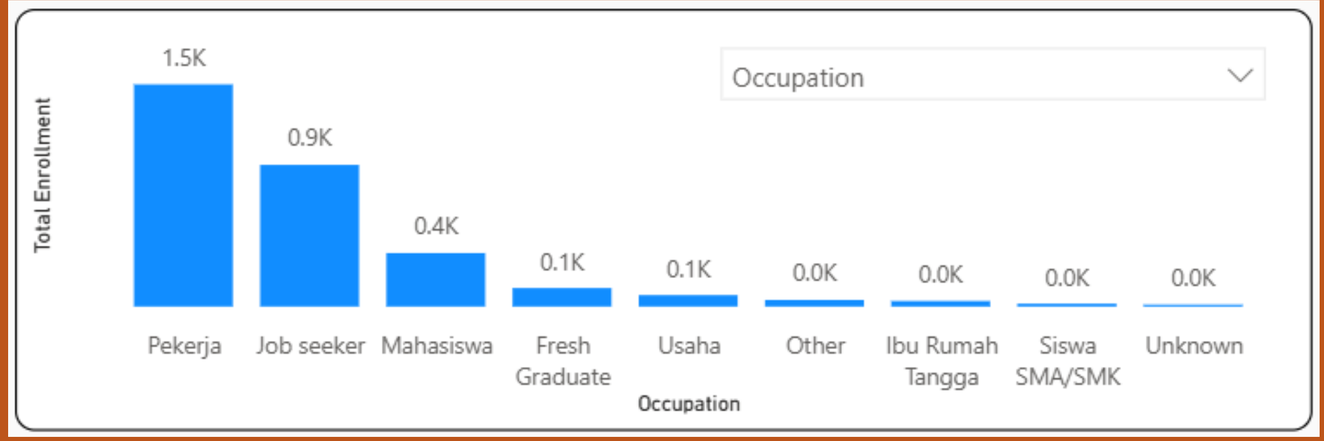
Business Vaulue

1. Improve visibility into real-time enrollment performance
2. Enable early detection of product demand deterioration
3. Forecast short-term demand (3 months)

Interactive Dashboard

OUR CORE MARKET IS EARLY-CAREER PROFESSIONALS



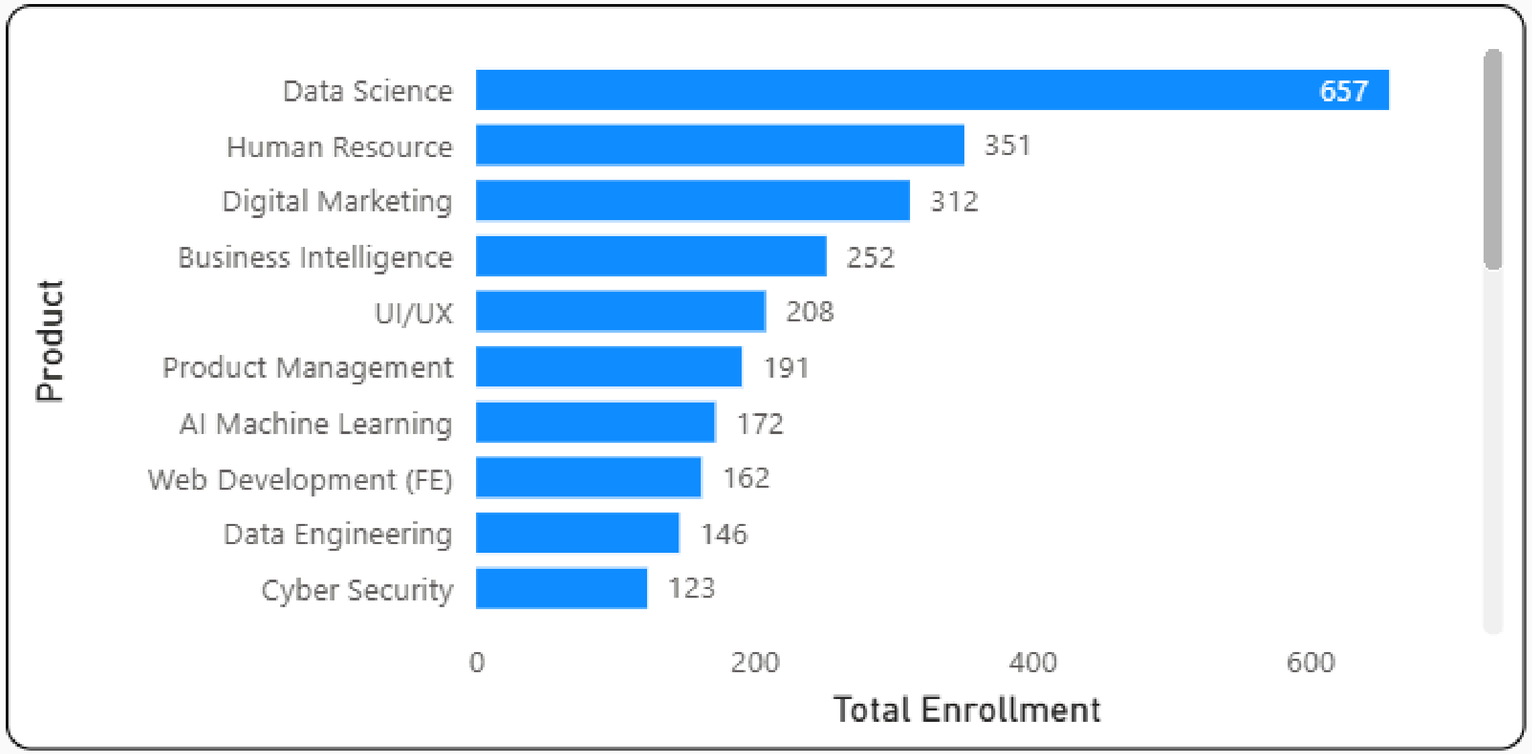


- The majority of enrollments come from individuals aged 20–30.
- Most participants are working professionals and job seekers.
- Demand is strongly driven by early-career skill upgrading needs.

Interactive Dashboard

ONE PRODUCT DRIVES THE BUSINESS

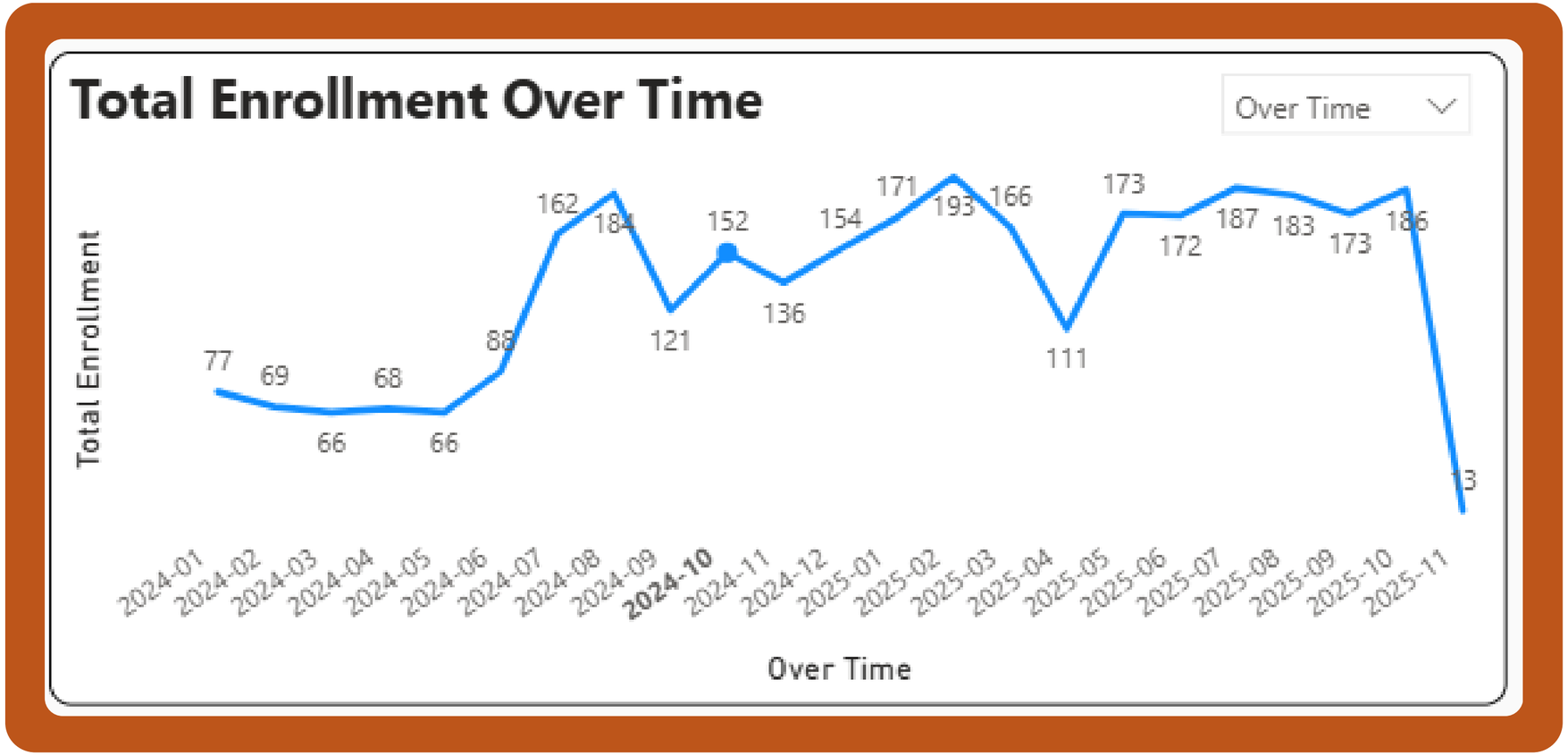
DATA SCIENCE LEADS THE PORTFOLIO



- Data Science (657) ≈ 2x Human Resource (351).
- Enrollment demand is highly concentrated in one flagship program.
- The portfolio shows imbalance, increasing dependency on top-performing products.

Interactive Dashboard

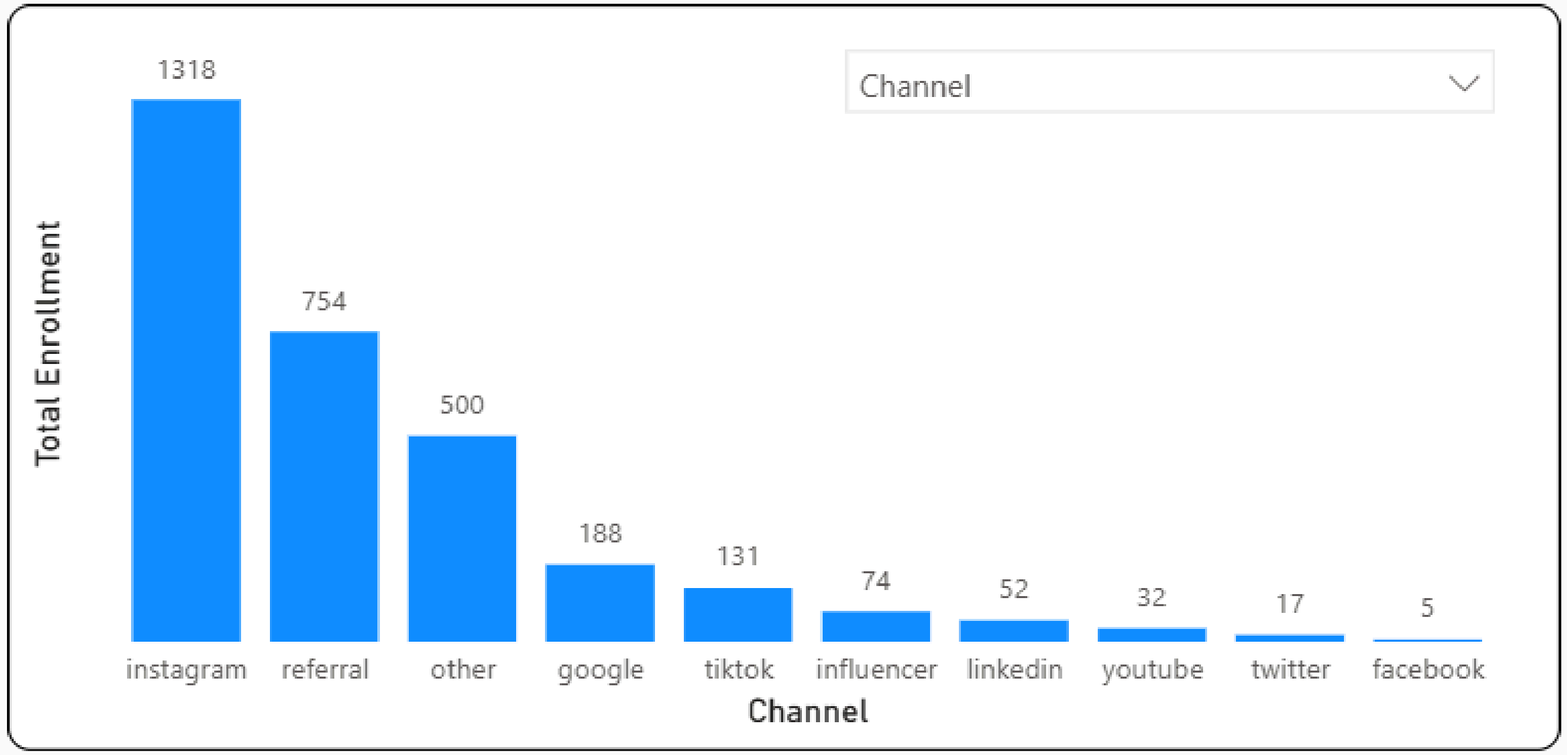
GROWTH IS VOLATILE, NOT SEASONAL



- A significant growth spike occurred around mid-2024.
- Enrollment throughout 2025 shows fluctuation rather than steady growth.
- No strong recurring seasonal pattern is observed.
- The last-month drop likely reflects incomplete data.

Interactive Dashboard

HIGH ACQUISITION DEPENDENCY ON

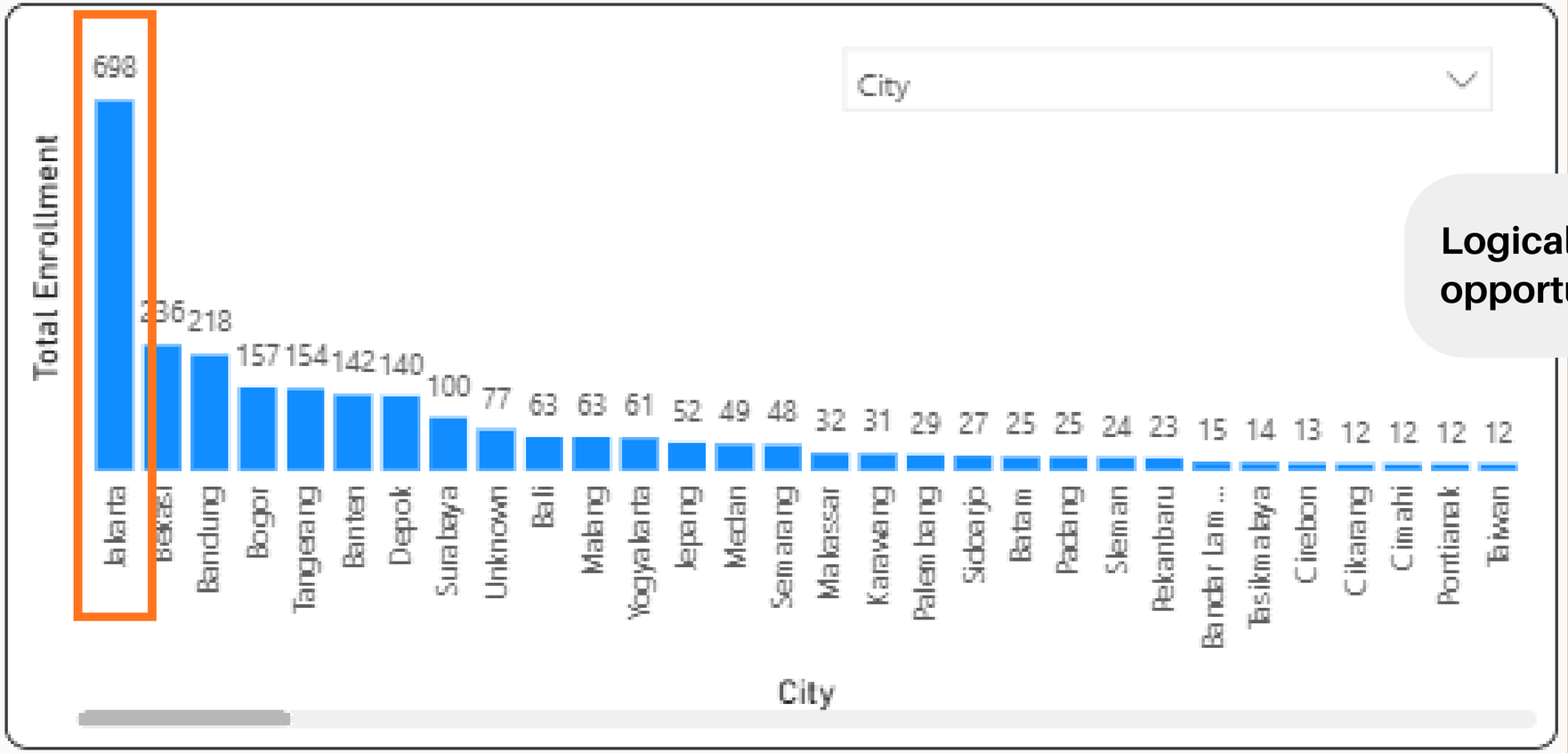


INSTAGRAM

- Instagram is the dominant acquisition channel.
- Referral is the second strongest contributor, indicating strong word-of-mouth effects.
- Heavy reliance on a single platform creates strategic risk exposure.

Interactive Dashboard

JAKARTA AS DEMAND HUB



Logical offline pilot opportunity

Interactive Dashboard

STUDENTS ENROLL FOR CAREER ACCELERATION

Motivation Cloud
bidang
bimbing
bootcamp
tarik
kerja
career
data
ajar
skill
pilih
harga
program
tujuan
tunjang
fasilitas
kembang
utama
silabus
marketing
job
connector
platform
rekomendasi
digital
science
analyst
kuliah
tambah
butuh
affordable
switch
teman
testimoni
lengkap
ilmu
depan
ikut
materi
kurikulum
business
coba
magang
ambil
cari
sesuai
bagus
peluang
hr
dalam
karir
usaha
kantor
berkarir
siap
security
kaya
engineering
banyak
opportunity
kampus
sebenarnya
project
udah
instagram
alumni
pindah
jalan
dicompare
engineer
goalsnya
fleksibel
online
management
improve
percaya
informasi
didik
salah
internship
jadwal
development
paham
detail
teknologi
careernya
motivasi
garansi
web
portofolio
hasil
cocok
tawar
langsung
dunia
kelas
sistem
proses
cv
karimya
kait
durasi
banding
hubung
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connect
intelligence
skillnya
turut
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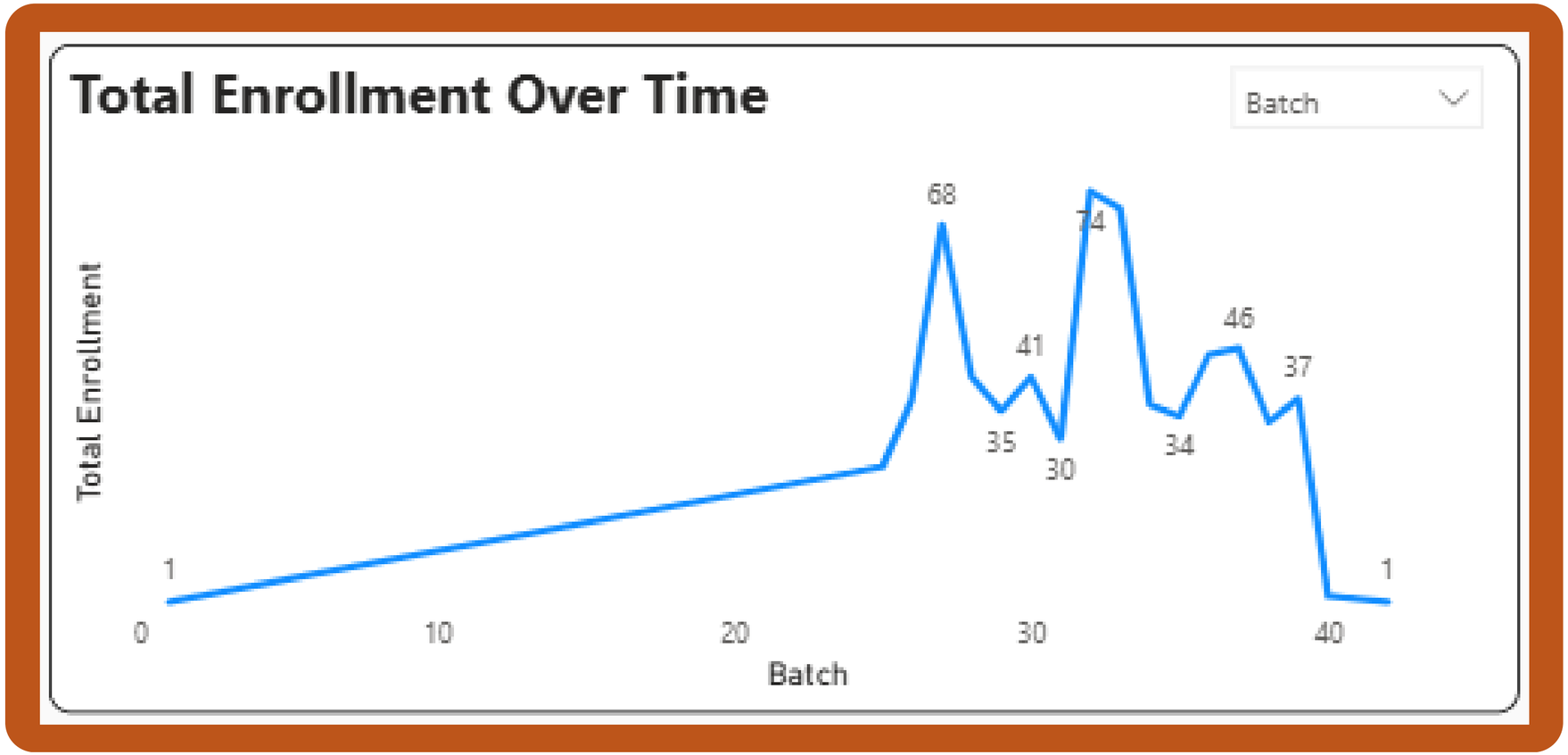
kait
teman

DATA BIAS & LIMITATION

Data Science Program



- Overlapping product labels (Data Science, DSA, AI/ML).
- Incomplete batch records for some products.

short-term forecasting and channel diversification are more critical than long-term seasonal modeling

Interactive Dashboard

Forcasting Model

Qualified Products:

	product	total_enrollment	distinct_months	zero_months	zero_ratio
8	Data Science	657	23	1	0.043478
17	Human Resource	351	23	4	0.173913
11	Digital Marketing	312	23	0	0.000000
4	Business Intelligence	252	23	1	0.043478
24	UI/UX	208	23	1	0.043478
20	Product Management	191	23	1	0.043478
0	AI Machine Learning	172	23	2	0.086957
25	Web Development (FE)	162	23	1	0.043478
7	Data Engineering	146	23	1	0.043478

link notebook

- Monthly enrollment aggregated per product
- Train–test split: 18 months train, 5 months test
- Forecast horizon: 3 months
- Evaluation metrics:
  - MAE (primary)
  - RMSE
  - SMAPE
- Model tested:
  - Based: Naïve, MA3
  - ARIMA (non-seasonal)
  - SARIMA
  - ETS

Model Evaluation Metrics

	product	model	MAE	RMSE	MAPE	SMAPE	Forecast_Quality
25	Web Development (FE)	MA3	2.600000	2.962731	inf	85.654928	High Volatility
23	UI/UX	ARIMA	3.233333	4.856776	inf	56.479622	High Volatility
6	Data Engineering	Naive	4.000000	4.732864	inf	73.142857	High Volatility
20	Product Management	ARIMA	4.022222	5.421266	inf	78.757311	High Volatility
12	Digital Marketing	Naive	4.400000	5.366563	64.121212	51.465933	High Volatility
4	Business Intelligence	MA3	5.000000	5.674504	inf	82.943723	High Volatility
2	AI Machine Learning	ARIMA	6.164361	7.075105	inf	113.698309	High Volatility
17	Human Resource	ARIMA	8.000000	9.402127	inf	80.257224	High Volatility
10	Data Science	MA3	22.600000	27.287360	inf	99.383937	High Volatility

link notebook

Short-term forecasting across 9 qualified products shows

three demand tiers

• High-demand programs

Forecast indicates stable demand above ~5 enrollments per month.

- Data Science
- AI / Machine Learning
- Business Intelligence
- Product Management

• Moderate-demand programs

Demand remains consistent but moderate.

- UI/UX
- Web Development (Front-End)

• Low-demand programs

Forecast suggests low but persistent demand, typically near 0–3 enrollments per month.

- Data Engineering
- Digital Marketing
- Human Resource

link notebook

INSIGHTS & RECOMMENDATIONS

KEY INSIGHT

- Enrollment demand is highly concentrated in a few flagship products, particularly Data Science.
- Demand shows volatility rather than strong seasonal patterns.
- Growth spikes appear campaign-driven, especially through Instagram and referral channels.
- Customer motivation is largely career advancement and skill improvement.
- Short-term forecasting across 9 qualified products shows three demand tiers.

Focus on flagship programs

Programs like Data Science and AI/ML should remain the primary growth drivers.

Align marketing with demand drivers

Given the strong role of campaigns and referral channels, marketing strategies should emphasize:

- referral incentives
- career-outcome messaging

THANK YOU FOR YOUR ATTENTION.

CONTACT

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GitHub Repository

DASHBOARD

Product Enrollment Performance Dashboard

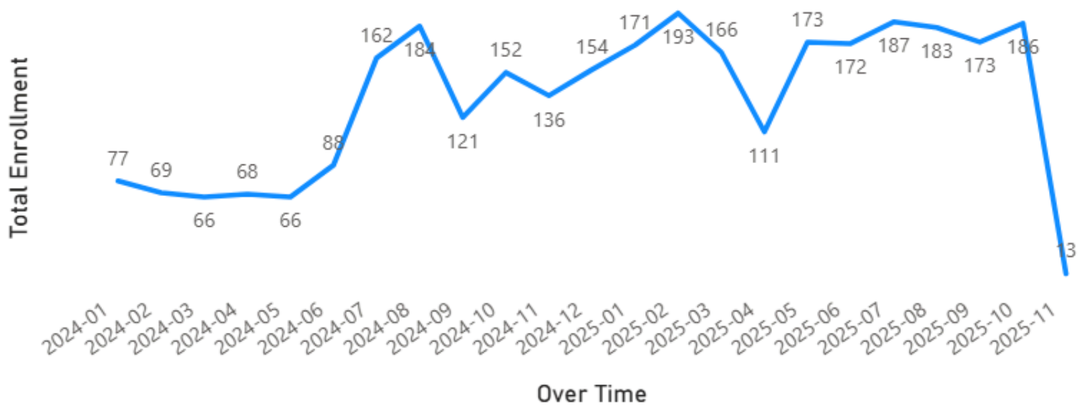
Product: All

Channel: All

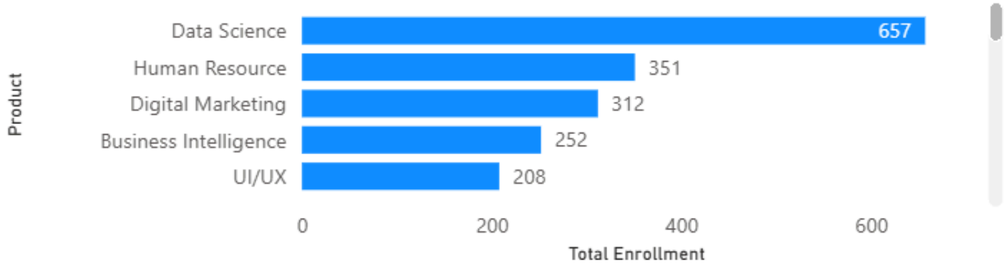
Active Products	Total Enrollment	Total Enrolled CAJC	Total Enrolled JCJC+
26	3071	1312	1759

Total Enrollment Over Time

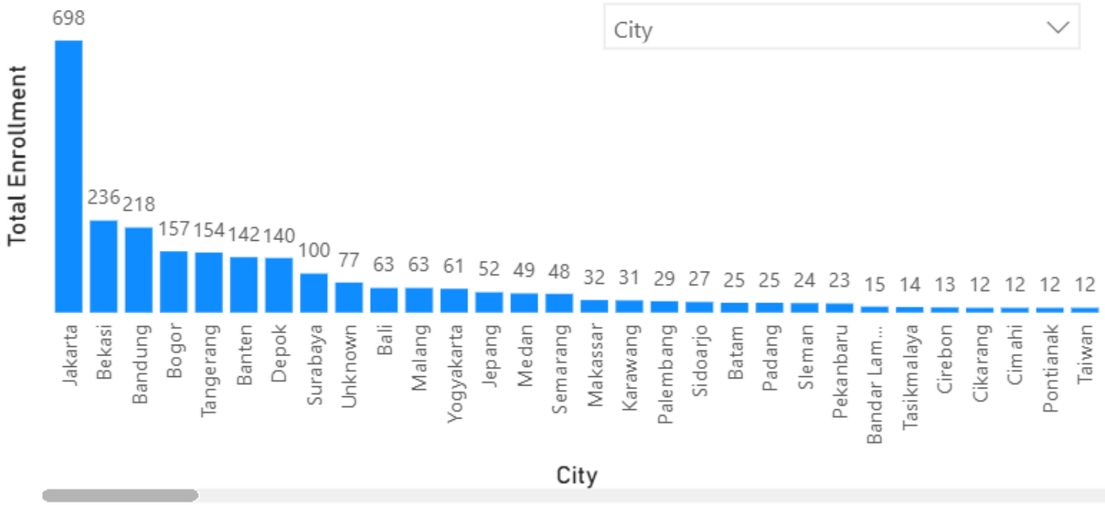
Over Time



Top 5 Product



City



Motivation Cloud

bimbing
bootcamp
bidang
kerja
data
career
tarik
skill
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connector
platform
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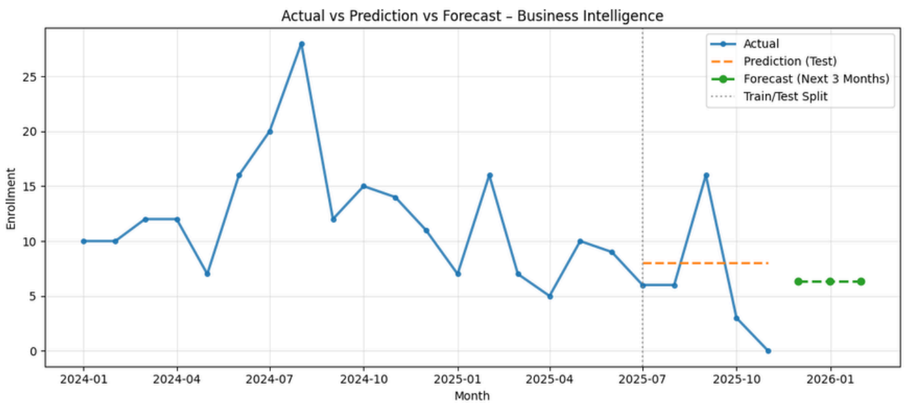
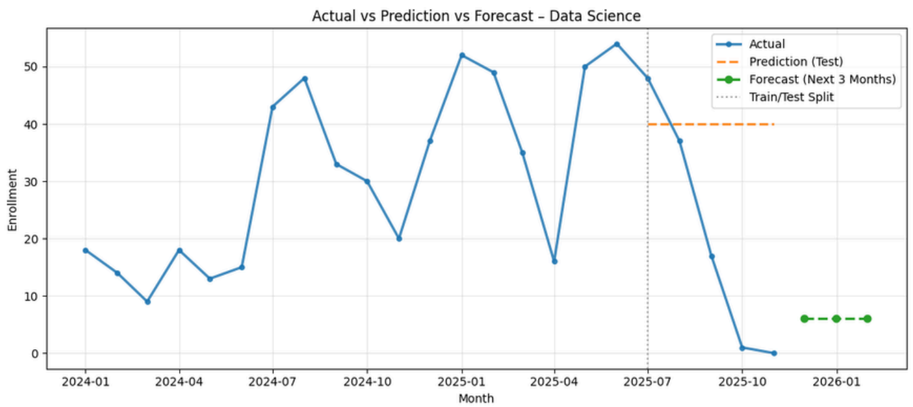
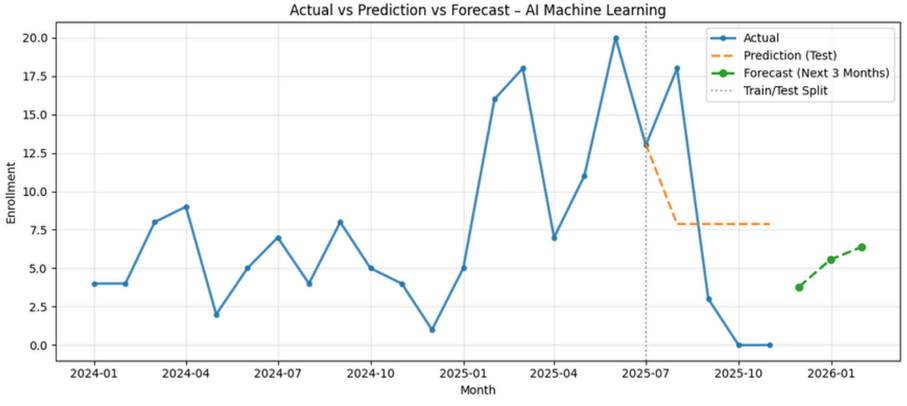
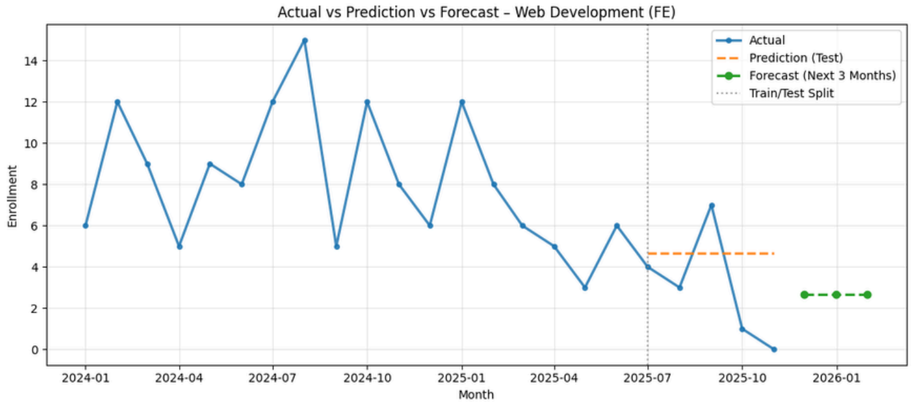
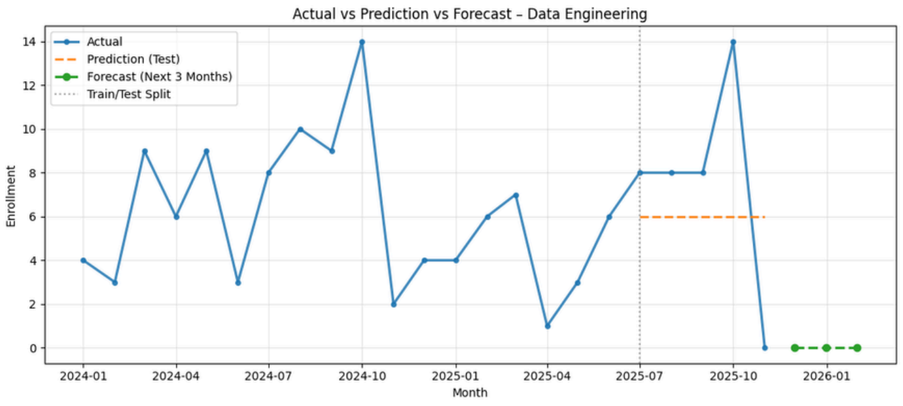
FORCASTING

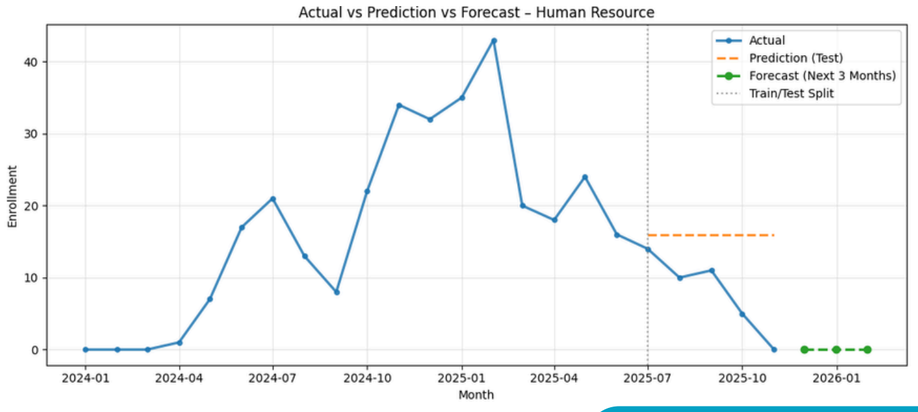












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